

PHYSICAL DESIGN DELIVERABLE

GDS II

GDS II is the final tapeout data that describes the physical layout of the chip using geometric shapes on different layers. It is the universal industry standard for IC fabrication.

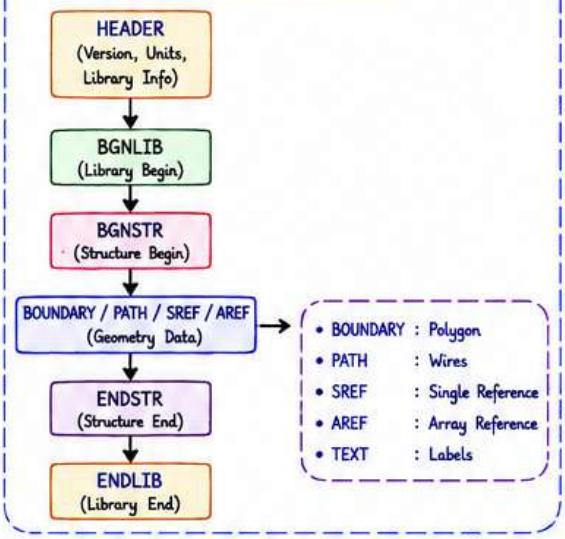
WHAT IS GDS II ?

- Binary file format used to describe physical layout data.
- Developed by Calma, now maintained by OASIS / Accellera.
- Stores shapes on layers with their properties.
- Used by all foundries for mask generation.

MAIN FEATURES

- ✓ Hierarchical data model
- ✓ Layer based representation
- ✓ Efficient data compression
- ✓ Technology independent
- ✓ Widely accepted by foundries

GDS II STRUCTURE



GDS II LAYERS EXAMPLE

LAYER	PURPOSE	DATATYPE (EXAMPLE)
1	NWELL	0
2	PWELL	0
10	ACTIVE	0
20	POLY	0
30	M1	0
⋮	⋮	⋮
200	PAD	0

(Layer numbers & datatypes are technology dependent)

WHY GDS II IS IMPORTANT ?

- ✓ Final representation of the chip.
- ✓ Input to mask shop / foundry.
- ✓ Ensures correct fabrication.
- ✓ Universally accepted format.
- ✓ Enables physical verification and analysis.

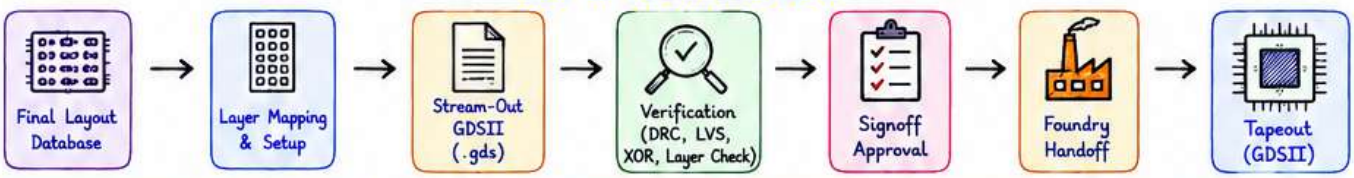
KEY POINTS

- ★ Use correct units & database precision.
- ★ Ensure consistent layer mapping.
- ★ Run complete verification before signoff.
- ★ Follow foundry GDSII requirements.
- ★ Clean, verified GDS = First pass success.

GOOD GDS II DELIVERS

- ✓ Clean DRC & LVS
- ✓ No opens / shorts
- ✓ Correct layer mapping
- ✓ Optimized file size
- ✓ Smooth foundry acceptance
- ✓ On-time tapeout ☺

GDS II FLOW OVERVIEW



COMMON FILE OPERATIONS

- ★ Stream-Out : Generate GDSII
- ★ Stream-In : Read GDSII
- ★ Layer Map : Tech ↔ GDS Layers
- ★ Merge / Split : Manage Libraries
- ★ Compress : Reduce File Size

VERIFICATION CHECKS

- ★ DRC (Calibre)
- ★ LVS (Calibre / Pegasus)
- ★ XOR (Calibre XOR / ICV)
- ★ Layer Check (Script / Manual)
- ★ Geometry Integrity

FOUNDRY HANDOFF PACKAGE

- ★ GDSII / OASIS file
- ★ Technology Files
- ★ Layer Map File
- ★ README / Release Notes
- ★ Signoff Reports

POST TAPEOUT

- ★ Mask data preparation
- ★ Wafer fabrication
- ★ Wafer sort & testing
- ★ Silicon bring-up
- ★ Characterization

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VDD

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ALL THE BEST !

Post - PD GDS II Generation and TapeOut

1. GDS II - INTRODUCTION

- GDS II (Graphic Data System - II) is the industry standard file format used to represent the final physical layout of an IC for fabrication.
- It contains all the geometric shapes, layers and their hierarchy information of the chip.
- Generated after post-route verification and sign-off.

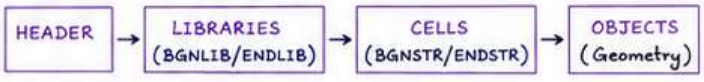
2. WHAT IS GDS II ?

- A binary file that describes the layout geometry.
- Used by mask data preparation (MDP) tools to create photomasks.
- Contains all mask layers, structures, text, cells, hierarchy and coordinates.
- Industry standard since 1980's.
- Latest version - GDS II (Calma format).

3. FEATURES

- Hierarchical data representation.
- Layer based geometry description.
- Compact binary format.
- Supports complex shapes (polygon, path etc.).
- Widely supported by EDA tools and foundries.
- Essential for mask generation and tapeout.

4. GDS II STRUCTURE OVERVIEW



Hierarchy : Library → Cell → Objects

5. GDS II OBJECTS

GDS II file is built using basic objects to define geometry and hierarchy.

OBJECT	DESCRIPTION	USED FOR
BOUNDARY	Defines closed polygon shape.	Block shapes, diffusion, well, active regions
PATH	Defines a path with width.	Wires, interconnects
TEXT	Stores text information.	Labels, pin names, annotations
SREF	Single Reference.	Instance of a cell at a location
AREF	Array Reference.	Array of cell instances
BOX	Rectangle defined by 2 diagonal points.	FP markings, keep-out regions etc.
NODE	Electrical node name.	Net names

6. POST - PD SIGN-OFF CHECKS (Before GDS)

- ✓ Timing Sign-off (Setup / Hold closed)
- ✓ DRC (Design Rule Check) - Clean.
- ✓ LVS (Layout vs Schematic) - Matched.
- ✓ ERC (Electrical Rule Check) - Clean.
- ✓ Antenna Check - Clean.
- ✓ IR Drop - Within limit.
- ✓ EM (Electromigration) - Clean.
- ✓ Metal Density - OK.
- ✓ Power Intent / UPF - Verified.
- ✓ TSV / Bump DRC - OK.

KEY POINT

GDS II is the final deliverable that represents the chip layout geometry. A clean GDS ensures smooth fabrication and high-first-pass silicon success.

7. VIVA / INTERVIEW POINTS

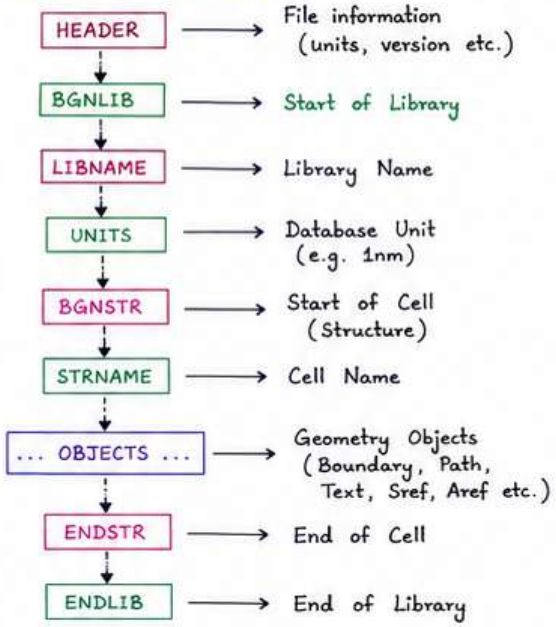
Q1. Why is GDS II important in Physical Design ?

→ It is the final layout database used for mask generation and fabrication. Any error in GDS can cause chip failure.

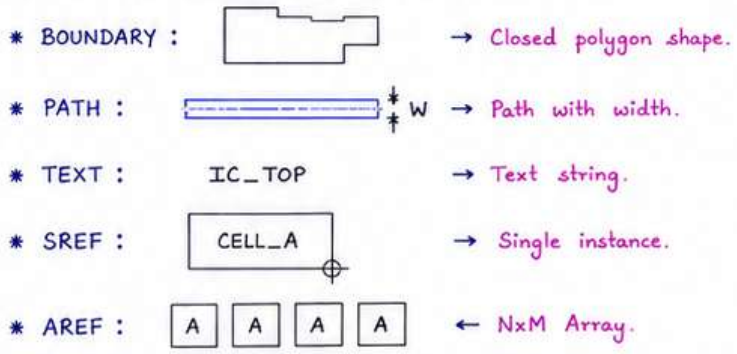
Q2. What checks are mandatory before GDS generation ?

→ Timing, DRC, LVS, ERC, Antenna, IR Drop, EM, Metal Density, Power Intent are mandatory.

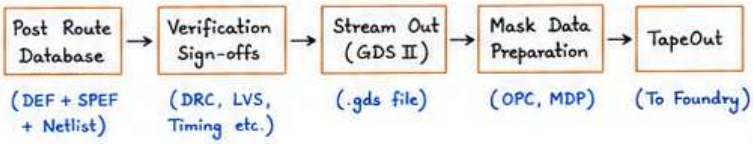
8. GDS II STRUCTURE (Detailed)



9. GDS II OBJECT REPRESENTATION



10. GDS II GENERATION FLOW



11. TAPEOUT CHECKLIST

- ✓ Final GDS reviewed.
- ✓ All sign-offs complete.
- ✓ Version control locked.
- ✓ Tapeout readiness review done.
- ✓ Foundry rule deck match.
- ✓ Engineering Change Order (ECO) closed.
- ✓ Final GDS archived.
- ✓ Release to Foundry.

12. TAPEOUT FLOW



GDS II is the starting point for fabrication.

13. SUMMARY / REVISION POINTS

- GDS II is the standard layout file used for tapeout.
- It stores geometry on different layers with hierarchy.
- Main objects - Boundary, Path, Text, Sref, Aref.
- GDS is generated after all sign-off checks are clean.
- MDP tools use GDS to create manufacturing masks.
- Clean GDS ensures smooth fabrication and better yield.
- Tapeout is the final milestone of Physical Design.

☺ ALL THE BEST !

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Post - PD GDS II Generation and TapeOut

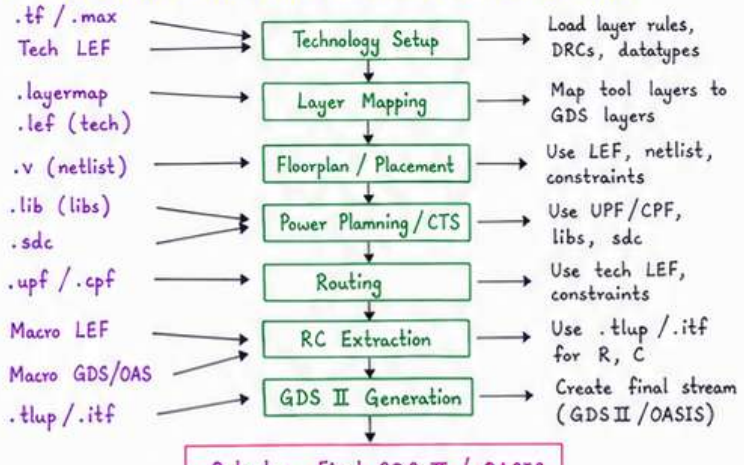
1. GDS II INPUTS - OVERVIEW

- GDS II generation requires multiple design, technology, constraint and library files produced during Physical Design.
- These inputs provide all the information required to create a correct, rule-clean and manufacturable layout.
- Quality of inputs directly affects the final GDS II quality.

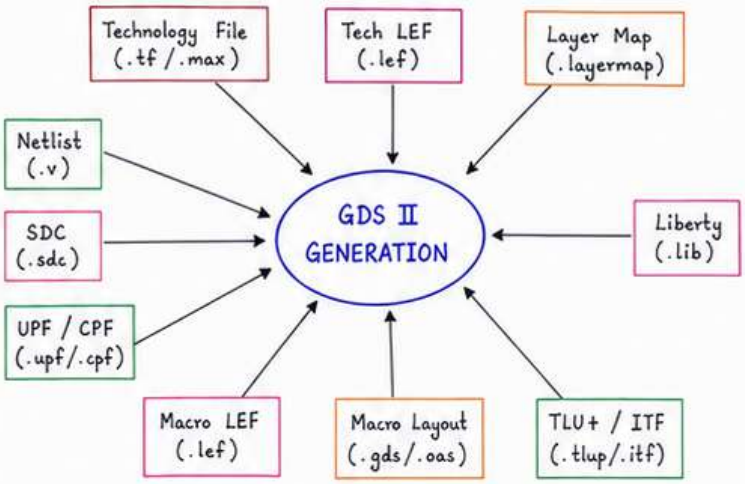
2. GDS II INPUTS - DETAILS

S.NO.	INPUT FILE	EXTENSION	DESCRIPTION / PURPOSE	PRODUCED BY
1.	Technology File (Process File)	.tf or .max	Contains layer definitions, datatypes, design rules, biasing, property info required for GDS creation.	Foundry
2.	Tech LEF (Technology LEF)	.lef	Defines routing layers, pitch, spacing, blockages, via rules, manufacturing constraints.	Foundry
3.	Layer Mapping File (Layer Map)	.layermap	Maps tool layer names to GDS II layer numbers (layer/datatype mapping).	PD Tool or Foundry
4.	Gate-Level Netlist	.v	Synthesized netlist in Verilog format. Contains instances and connections.	Logic Synthesis Team
5.	Design Constraints (SDC)	.sdc	Synopsys Design Constraints file. Contains clocks, I/O, timing, false paths, etc.	STA / PD Team
6.	Power Intent File (UPF / CPF)	.upf or .cpf	Defines power domains, power states, isolation, retention, level shifters and power strategies.	PD / Power Team
7.	Macro LEF (Macro Abstracts)	.lef	Abstract views of hard macros / IPs. Defines pins, blockages, obs, etc.	IP Vendor/ PD Team
8.	Macro Layout (GDS / OAS)	.gds or .oas	Full layout of macros/IPs used in the design. Inserted during top-level integration.	IP Vendor
9.	Liberty Files (Timing Libraries)	.lib	Cell timing, power, area information for STA and optimization.	Library / Foundry
10.	Parasitic Tech File (TLU+)	.tlup or .itf	Technology dependent R, C tables and layer capacitance information used for extraction.	Foundry
11.	(Optional) Simulation Netlist	.v	Post-Layout netlist (SDF annotated) used for gate-level simulation.	PD / Verification Team

3. HOW INPUTS ARE USED IN GDS II GENERATION FLOW



4. INPUT FILE RELATIONSHIP DIAGRAM



5. INPUT FILE CATEGORIES

- ✱ Technology Inputs : .tf/.max, Tech LEF, .layermap, .tlup/.itf
- ✱ Design Inputs : .v (netlist), Macro LEF, Macro GDS/OAS
- ✱ Constraint Inputs : .sdc, .upf/.cpf
- ✱ Library Inputs : .lib

6. IMPORTANCE OF EACH INPUT

- ✓ .tf/.max = Defines process and layer rules.
- ✓ Tech LEF = Defines routing layers, vias, spacing, DRC rules.
- ✓ .layermap = Ensures correct mapping between tool and GDS.
- ✓ .v (netlist) = Provides logical connectivity of the design.
- ✓ .sdc = Ensures timing requirements are met.
- ✓ .upf/.cpf = Ensures correct power architecture in layout.
- ✓ Macro LEF = Provides abstract info for macros.
- ✓ Macro GDS/OAS = Provides actual layout for macros.
- ✓ .lib = Needed for timing, power and optimization.
- ✓ .tlup/.itf = Critical for accurate RC extraction.

KEY POINT

All the above inputs must be consistent and from the same technology and revision to ensure a correct, rule-clean GDS II.

7. COMMON INPUT CHECKS BEFORE GDS GENERATION

- ✓ Technology file version check
- ✓ Layer map correctness
- ✓ LEF syntax & DRC check
- ✓ Netlist connectivity check
- ✓ SDC syntax & clock check
- ✓ UPF/CPF power intent check
- ✓ Macro LEF pin alignment
- ✓ Macro layout vs LEF match
- ✓ Liberty file completeness
- ✓ TLU+ tech file compatibility

8. PRACTICAL OBSERVATIONS

- → Wrong layer mapping leads to incorrect layers in GDS.
- → Mismatch between Macro LEF and GDS causes DRC/LVS issues.
- → Old technology files can cause false DRC violations.
- → Missing or wrong SDC leads to timing failure in final chip.

9. VIVA / INTERVIEW POINTS

Q1. What are the main inputs required for GDS II generation ?
→ Technology files, LEF files, netlist, constraints, power intent, macro files, libraries and parasitic tech files.

Q2. Why is layer mapping file important in GDS generation ?
→ It maps tool layer names to GDS layer numbers and ensures correct layer/datatype in the final GDS II.

ALL THE BEST !

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Post - PD GDS II Generation and TapeOut

3. STREAM-OUT PROCESS

- **Definition :**

  Stream-out (also called database write) is the process of writing the final verified layout database from the Physical Design tool to a manufacturing file format (GDS II or OASIS) using Synopsys Fusion Compiler (FC). The stream file contains all polygons, vias, text, instances, and hierarchy needed for mask generation.

- **Purpose :**
  - ✓ To create a standard transfer format for foundry.
  - ✓ Enables mask data preparation (MDP).
  - ✓ Acts as the final handoff for fabrication.
  - ✓ Preserves full hierarchy and layer mapping.

- **Typical Stage in Flow :**

Physical Verification → Sign-off Checks → Database Cleanup → Stream-Out (GDS II/OASIS) → Mask Data Prep → TapeOut (Foundry)

(This Stage)

4. STREAM-OUT COMMANDS IN SYNOPSYS FC

The main command used in Fusion Compiler to write GDSII or OASIS is :

```
write_stream -format {gds | oasis} -output <file> [options]
```

Examples :

- `write_stream -format gds -output chip_top.gds`
- `write_stream -format oasis -output chip_top.oas`

Supporting Commands :

- `check_stream` - verify database before writing.
- `report_stream` - summary of the stream file to be written.
- `specify_layers` - define layer mapping (if not in techfile).
- `set_stream_options` - configure stream-out options.

5. COMMON PROPERTIES / OPTIONS IN FC STREAM-OUT

OPTION	SYNTAX / EXAMPLE	DESCRIPTION	DEFAULT
-format	-format {gds | oasis}	Choose output format. GDS II or OASIS.	gds
-output	-output chip.gds	Output file name.	None (required)
-units	-units 1000 (or DBU)	Database units per micron (dbu).	Tool DBU (usually 2000)
-hierarchy	-hierarchy {preserve | none | flatten}	How hierarchy is written in stream.	preserve
-techfile	-techfile tech.tf	Technology file (.tf/.tf or .map) for layer mapping & stream rules.	Loaded tech file
-map_file	-map_file layer.map	Layer mapping file (if separate).	None
-compress	-compress {on | off}	Compress output (GDSII/OASIS).	on
-mode	-mode {ALL | TOP | SELECTED}	Stream mode. ALL or specific cells.	ALL
-bbox	-bbox {xmin ymin xmax ymax}	Write only within bounding box.	Full chip area
-precision	-precision {1 | 2 | 4}	Coordinate precision (1nm/10nm etc).	2
-via_rule	-via_rule {expanded | collapsed}	How vias are written (expanded polygons or collapsed holes).	expanded
-merge_polygons	-merge_polygons {on | off}	Merge touching polygons for cleaner stream.	on

6. DETAILED OPTIONS WITH EXPLANATIONS

- ✱ -format : Select GDS II or OASIS.
- ✱ -units : Define db units per micron. (e.g., 1000 = 1nm)
- ✱ -hierarchy :
  - preserve → keep full hierarchy (recommended)
  - flatten → write single cell (no hierarchy)
  - none → write top level only
- ✱ -techfile : Provides layer datatypes, stream rules, min/cut sizes, etc.
- ✱ -compress : Reduces file size.
- ✱ -bbox : Write only a region (useful for blocks).
- ✱ -via_rule : Expanded → via as rectangle + hole
  Collapsed → via as single hole.
- ✱ -merge_polygons : Removes redundant edges and merges to reduce file size.
- ✱ -mode : ALL → all cells in design.
  SELECTED → user specified cells.
- ✱ -precision : Coordinate rounding precision.
- ✱ -timestamp : Add time stamp in stream header.
- ✱ -property_file : Attach user properties in GDSII.
- ✱ -append : Append to an existing stream (rare).

7. TYPICAL STREAM-OUT FLOW IN FC

Run sign-off checks (DRC, LVS, Antenna, STA etc.)	→ Ensure 0 critical violations
↓	
Database cleanup & ECO (Fix violations)	→ Remove fillers/markers if required
↓	
Set stream options (set_stream_options)	→ Set units, hierarchy, techfile, compress etc.
↓	
Check stream (check_stream)	→ Verify design DB before writing
↓	
Write stream (write_stream -format gds/oasis)	→ Generate GDS II / OASIS file
↓	
Report generation (report_stream)	→ Check file size, cell count, layer count etc.
↓	
Deliver to MDP/Foundry	→ MDP → Mask data → Fabrication

8. SAMPLE FC SCRIPT FOR STREAM-OUT (GDS II)

```
# Set technology file
set_technology -tf mytech.tf                    ; # techfile with layers datatypes & rules
# Set stream options
set_stream_options -units 1000                  ; # database units (1nm)
set_stream_options -hierarchy preserve          ; # keep hierarchy
set_stream_options -compress on                 ; # compress output
set_stream_options -merge_polygons on           ; # clean polygons
# Check design before stream-out
check_stream -verbose
# Write GDSII file
write_stream -format gds -output chip_top.gds
# Report
report_stream -output stream_report.rpt         ; # summary report
```

9. BEST PRACTICES

- ✓ Always run full sign-off checks before stream-out.
- ✓ Use preserve hierarchy unless foundry requires flat.
- ✓ Use correct technology file and layer map.
- ✓ Compress output to reduce file size.
- ✓ Validate stream (check_stream & report_stream).
- ✓ Coordinate db units with foundry requirement.

KEY POINT

Stream-out is the final step in PD that converts the internal database into a standard manufacturable file (GDS II/OASIS). Clean, rule-correct stream ensures smooth tape-out and better yield.

10. VIVA / INTERVIEW POINTS

Q1. What is the purpose of stream-out in Physical Design ?
→ To generate final layout file (GDS II/OASIS) for mask data preparation and fabrication.

Q2. Why is technology file important during stream-out ?
→ It provides layer datatypes, stream rules, min/cut sizes and mapping to write a foundry-acceptable stream.

Q3. Difference between -hierarchy preserve and flatten ?
→ Preserve keeps cell hierarchy; Flatten writes a single top cell (large file, not preferred).

ALL THE BEST !

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Post - PD GDS II Generation and TapeOut

1. OASIS FORMAT - INTRODUCTION

- OASIS (Open Artwork System Interchange Standard) is an advanced layout file format developed by Si2 for efficient storage and transfer of IC layout data.
- It is an alternative to GDSII and is widely used in modern VLSI Physical Design for large and complex SoCs.
- Provides significantly smaller file sizes compared to GDSII using advanced compression techniques.

2. OASIS vs GDS II

Parameter	GDS II	OASIS
Full Form	Graphic Data System	Open Artwork System Interchange Standard
File Size	Large	Much smaller (compressed)
Technology	Older (1980s)	Modern (1998+)
Data Encoding	Fixed length records	Variable length records
Hierarchy	Basic support	Better & efficient support
Repeats	Not efficient	Highly efficient (repetition)
Usage	Still used, legacy	Preferred for advanced nodes / large SoCs

KEY POINT

OASIS provides same layout information as GDSII but with much smaller file size through intelligent compression techniques.

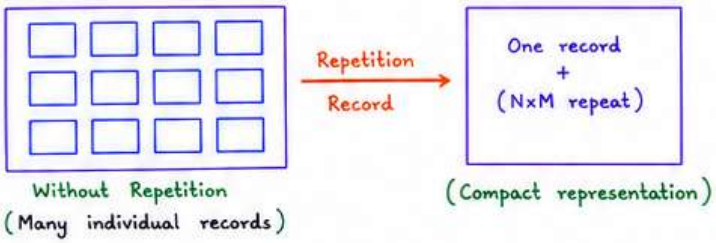
3. NEED FOR OASIS

- ✓ Modern SoCs have billions of polygons → GDSII becomes huge.
- ✓ Faster data transfer and reduced storage cost.
- ✓ Efficient representation of repeated and regular structures.
- ✓ Better support for hierarchical and sparse data.
- ✓ Widely adopted in advanced technology nodes (7nm, 5nm, 3nm...).

4. COMPRESSION TECHNIQUES IN OASIS

(i) Repetition Records

- Stores multiple identical geometries using a single record.
- Specifies one geometry and its repetition (rows x columns).
- Reduces file size for regular structures (e.g., std. cell rows).



Without Repetition
(Many individual records)

(Compact representation)

(ii) Delta Encoding

- Stores only the difference (delta) between successive coordinates instead of absolute values.
- Consecutive points are close → deltas are small → fewer bits.
- Very effective for long paths and polygon vertices.

Example :

Absolute :	(1000, 1000)	(1010, 1005)	(1020, 1010)
		↓	↓
Delta :	(1000, 1000)	(+10, +5)	(+10, +5)

(iii) Modal Variables

- Uses previously used values as default (modal) for next records.
- Only changes are written, not the entire data again.
- Commonly used for : layer number, datatype, text, width etc.
- Reduces redundant information and total file size.

Example :

Instead of writing → Layer = 20, Datatype = 0 for every shape OASIS writes it once, and next shapes assume same values ntil it changes (modal behavior).

5. OTHER EFFICIENCY FEATURES

- ✱ Variable length records → no fixed 2-byte structure.
- ✱ Sparse data representation → skips empty areas.
- ✱ Efficient encoding of text, properties and cell names.
- ✱ Better support for very large hierarchical designs.

6. FILE SIZE COMPARISON (TYPICAL)

Design Size	GDS II	OASIS	Reduction
Small block	100 MB	10 - 20 MB	~ 5 - 8x
Large SoC	10 GB	0.5 - 1 GB	~ 10 - 20x
Advanced node	100 GB+	5 - 10 GB	~ 10x or more

7. OASIS SUPPORT IN TOOLS

- Synopsys Fusion Compiler : write_stream -format oasis
- Cadence Innovus : streamout -oas
- Mentor Calibre / KLayout : native OASIS support

8. USE IN TAPEOUT FLOW

Sign-off Layout → OASIS Stream-Out → PV (DRC/LVS) with OASIS → Foundry Submission

- ✓ Many leading foundries accept OASIS as the preferred format.
- ✓ Reduces data handling time and improves turnaround.
- ✓ Some flows may require both GDSII and OASIS for compatibility.

9. ADVANTAGES / LIMITATIONS

Advantages

- ✓ 5-20x smaller file size
- ✓ Faster I/O and transfer
- ✓ Efficient for large SoCs
- ✓ Better compression methods
- ✓ Modern tool & foundry support

Limitations

- Older tools may not support
- Slightly more complex format
- Debugging by manual view can be harder than GDSII

10. VIVA / INTERVIEW POINTS

Q1. Why is OASIS preferred over GDSII ?
→ Because it provides same layout information with significantly smaller file size using advanced compression.

Q2. What are the main compression techniques in OASIS ?
→ Repetition records, Delta encoding and Modal variables.

11. SUMMARY

- OASIS is a modern, compressed layout file format.
- Uses repetition records, delta encoding and modal variables.
- Achieves 5-20x smaller file size compared to GDSII.
- Widely supported by EDA tools and leading foundries.
- Preferred format for large and advanced VLSI designs.

ALL THE BEST !

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Post - PD GDS II Generation and TapeOut

7. GDS VERIFICATION - INTRODUCTION

- GDS Verification ensures that the final stream-out layout is geometrically correct, complete, consistent and ready for fabrication.
- It compares the written database with the in-memory layout and checks for any data corruption or missing information.
- Verification is a mandatory step before tapeout.

8. GDS VERIFICATION - STAGES (FLOW)

① Stream Out GDS / OASIS (Fusion Compiler) → ② Stream In Back (FC / Virtuoso) → ③ XOR Compare (Calibre XOR / ICV) → ④ Layer Check (Script / Manual) → ⑤ Final Validation (Pass / Fail) → ⑥ Geometric Integrity (Calibre DRC)

① Stream Out GDS : Generate final GDS II or OASIS from FC.

② Stream In Back : Read back the GDS/OASIS to a new design database (FC / Virtuoso).

③ XOR Compare : Perform bit-by-bit comparison between original in-memory design and streamed-in design using Calibre XOR or ICV.
Output = 0 differences.

④ Layer Check : Verify layer mapping, layer purpose, datatype, visibility, text, properties using script or manual inspection.

⑤ Final Validation (Pass / Fail) : If all checks are clean → PASS
else → FAIL and fix issues.

⑥ Geometric Integrity (Calibre DRC) : Run Calibre DRC on the streamed-in layout to ensure geometric correctness.

9. DETAILED DESCRIPTION OF STAGES

STAGE	PURPOSE	INPUT	OUTPUT	TOOLS	CHECKS / NOTES
① Stream Out GDS/OASIS	Write final layout to file	Final design database (FC)	.gds / .oas	Synopsys FC	Use correct techfile, layer map & stream options.
② Stream In Back	Read the streamed file back	.gds / .oas	New design database	FC / Virtuoso	Must load same techfile & layer map.
③ XOR Compare	Ensure no data loss / corruption	Original DB & Streamed-in DB	XOR DB & Report	Calibre XOR / ICV	0 XOR count → Clean.
④ Layer Check	Verify layer mapping & attributes	Streamed-in DB	Layer check Report	Script / Manual	Check purpose, datatype, text, visibility.
⑤ Final Validation	Consolidate all reports and decide	XOR report, Layer check report	Pass / Fail	Manual	All clear → Pass Else → Fail
⑥ Geom. Integrity (Calibre DRC)	Check geo. correctness of layout	Streamed-in DB	DRC report (Errors / 0)	Calibre DRC	All DRC clean → Ready for TapeOut.

KEY POINT

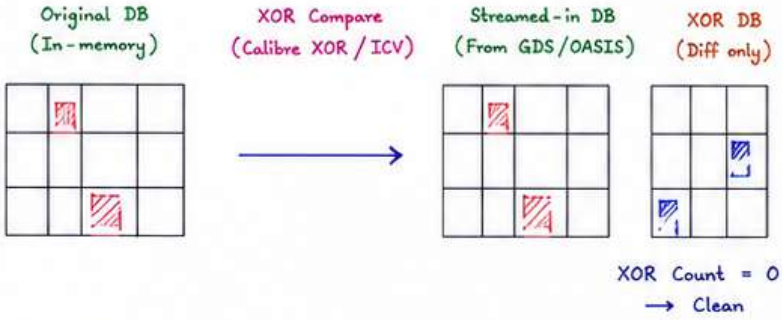
GDS Verification is the final quality gate before tapeout. Any issue found after tapeout can cause huge cost & delay. Hence "VERIFY → VERIFY → VERIFY".

10. GEOMETRIC INTEGRITY (CALIBRE DRC)

- Run DRC on the streamed-in database.
- Ensures layout follows all foundry design rules.
- Checks include : width, spacing, enclosure, overlap, notch, density, slot, antenna, min-area, etc.
- DRC must be 100% clean for tapeout sign-off.

11. XOR COMPARE - HOW IT WORKS

- Two databases are compared cell by cell and layer by layer.
- Differences are stored in XOR database.
- XOR count (number of mismatches) should be ZERO.



12. LAYER CHECKS - WHAT TO VERIFY

* Layer Mapping : Correct layer ↔ purpose ↔ datatype.
* Layer Presence : No missing or extra layers.
* Text : All labels, pin names, properties.
* Properties : Cellname, texttype, width, layer, etc.
* Datatypes : As per foundry requirement.
* Visibility : Hidden / visible layers as intended.
* Units : Database units & physical units.
* Hierarchy : Cell hierarchy intact.

Tip : Use layer check script or manual inspection in Virtuoso.

13. COMMON XOR / LAYER CHECK ISSUES

ISSUE	CAUSE	EFFECT	ACTION
Non-zero XOR count	Stream-out option mismatch	Data loss / Mismatch	Check stream options
Layer missing	Layer map error	Missing shapes	Verify layer map
Extra layer present	Wrong techfile	DRC/LVS fail	Load correct techfile
Wrong datatype	Layer mapping incorrect	Functional incorrectness	Fix mapping
Text missing	Text layers not streamed	Pin / label missing	Enable text stream-out

14. GDS VERIFICATION CHECKLIST

- ✓ Correct techfile used.
- ✓ Correct layer map used.
- ✓ Stream-out options reviewed.
- ✓ No warnings during stream-out.
- ✓ Stream-in successful.
- ✓ XOR compare clean (0 count).
- ✓ Layer check clean.
- ✓ Text and properties verified.
- ✓ Database units verified.
- ✓ Calibre DRC clean.
- ✓ All reports reviewed.
- ✓ Final sign-off approval.

15. SUMMARY / REVISION POINTS

- GDS Verification is essential to ensure final layout quality.
- Follow all verification stages in sequence.
- XOR compare must be zero for a clean verification.
- Layer check validates mapping, purpose, datatype and text.
- Calibre DRC confirms geometric integrity.
- Only after all checks PASS, the design is ready for tapeout.

16. VIVA / INTERVIEW POINTS

Q1. Why is GDS verification important before tapeout ?
→ To ensure no data loss/corruption and to guarantee geometric correctness of the final layout for manufacturing.

Q2. What tool is used for XOR comparison of GDS ?
→ Calibre XOR or ICV (IC Validator).

Q3. What does geometric integrity check ensure ?
→ It ensures the layout follows all foundry design rules using Calibre DRC.

Author : Konda Tharun Vivekananda

☺ VERIFY → SIGN-OFF → TAPEOUT → SILICON

Post - PD GDS II Generation and TapeOut

17. TAPEOUT DELIVERABLES - INTRODUCTION

- Tapeout deliverables are the final set of files and reports submitted to the foundry for manufacturing.
- These deliverables contain all the information required to fabricate the chip exactly as verified.
- All files must be consistent, sign-off clean and must meet foundry requirements.
- Deliverables are packaged, documented and handed over with release notes and checklist.

18. TAPEOUT DELIVERABLES - LIST

S.NO.	DELIVERABLE	DESCRIPTION / PURPOSE	FILE FORMAT/ EXTN.
1.	Layout (GDSII / OASIS)	Final mask data of the design. Contains all layers, shapes, instances, text, etc.	.gds .oas
2.	Netlist	Circuit connectivity (Flattened or Hierarchical) used for LVS and documentation.	.v .sp .sv
3.	SPEF	Parasitic RC extraction file for post-layout analysis and sign-off.	.spef
4.	SDC	Design constraints (clocks, I/O delays, false paths, etc.) used for STA.	.sdc
5.	Reports	Complete set of sign-off reports proving design quality and correctness.	.rpt .html .log
6.	Release Notes	Summary of design, process, tools, assumptions, ECOs, waivers, special notes.	.pdf .doc .txt
7.	Signoff Checklist	Final checklist showing all verifications completed and design sign-off status.	.xls .xlsx .pdf
8.	Layer Map	Mapping between GDS/OASIS layers and foundry layer names.	.layermap
9.	Other Supporting Files	Tech files, LEFs, Liberty, Power intent, UPF, CDL, DFT files, etc. (as reqd.).	various

KEY POINT

All deliverables must be consistent with each other and with the foundry data package. Any mismatch can cause tapeout rejection or silicon failure.

19. DELIVERABLES FLOW

Final Sign-off Clean Design → Generate Deliverables → Package & Verify → Submit to Foundry

20. BRIEF DESCRIPTION OF EACH DELIVERABLE

1. **Layout (GDSII / OASIS)** : Final geometrical database containing all layers, polygons, paths, text, references, etc. Ready for mask generation.
2. **Netlist** : Describes electrical connectivity of the design after synthesis and ECOs. Used for LVS and documentation.
3. **SPEF** : Standard Parasitic Exchange Format containing RC parasitics extracted from the final routed design.
4. **SDC** : Synopsys Design Constraints file with all timing, power and I/O constraints required for STA.
5. **Reports** : Proof of sign-off for all checks (DRC/LVS/STA/IR/EM/Antenna/Power, etc.).
6. **Release Notes** : Provides design overview, assumptions, tool versions, ECO list, waivers, special handling instructions.
7. **Signoff Checklist** : Shows Pass/Fail status of every sign-off item with tool, version and date.
8. **Layer Map** : Ensures correct interpretation of layers by the foundry tools.
9. **Other Files** : Additional files required for foundry and internal use (LEF, Liberty, UPF, CDL, DFT, etc.).

21. DELIVERABLES SUMMARY TABLE

DELIVERABLE	USED FOR	GENERATED BY	CHECKED BY
Layout (GDSII/OASIS)	Mask fabrication	Stream-out (FC)	DRC, Layer Check
Netlist	LVS, Documentation	Synthesis / ECO	LVS
SPEF	Post-layout STA	RC Extraction	Parasitic Sanity
SDC	STA Constraints	Constraint Team	STA
Reports	Sign-off Evidence	All Verification Tools	Design Team
Release Notes	Design Information	PD Leads	Mgmt / QA
Signoff Checklist	Tapeout Approval	PD Verification	QA / Leads
Layer Map	Layer Interpretation	PD / Tech Team	Layer Check
Other Files	Supporting Data	Various Teams	As Required

22. TAPEOUT DELIVERABLES PACKAGE STRUCTURE (EXAMPLE)

```
<DESIGN_NAME>_TAPEOUT_PACKAGE/
├── 01_LAYOUT/
│   ├── • design_final.gds
│   └── • design_final.oas
├── 02_NETLIST/
│   ├── • design.v
│   └── • top_flattened.v
├── 03_SPEF/
│   └── • design_final.spef
├── 04_CONSTRAINTS/
│   └── • design.sdc
├── 05_REPORTS/
│   ├── • DRC/
│   ├── • LVS/
│   ├── • STA/
│   ├── • IR_DROP/
│   ├── • EM/
│   ├── • ANTENNA/
│   └── • POWER/
├── 06_RELEASE_NOTES/
│   └── • release_notes.pdf
├── 07.SIGNOFF_CHECKLIST/
│   └── • signoff_checklist.xlsx
├── 08_LAYER_MAP/
│   ├── • layer.map
│   └── • layermap.lmp
├── 09/MISC/
│   ├── • techfile.tf
│   ├── • upf_file.upf
│   ├── • lef/
│   ├── • liberty/
│   ├── • cdl/
│   ├── • dft/
│   └── • others/
└── README.txt
```

Note : Folder names and file list may vary as per foundry requirements.

23. IMPORTANCE OF TAPEOUT DELIVERABLES

- ✓ Ensures the foundry receives all required data for mask generation.
- ✓ Provides traceability and documentation for the design.
- ✓ Helps in debugging if any silicon issue occurs.
- ✓ Demonstrates that design has passed all sign-off checks.
- ✓ Avoids schedule slip due to missing or incorrect files.

24. BEST PRACTICES

- ✓ Always verify all deliverables before packaging.
- ✓ Maintain naming convention and directory structure.
- ✓ Cross-check consistency between GDS, netlist and reports.
- ✓ Ensure all reports are clean with zero critical violations.
- ✓ Follow foundry tapeout manual strictly.
- ✓ Keep backup of the final package.

25. VIVA / INTERVIEW POINTS

Q1. What are tapeout deliverables ?

→ They are the final set of files (layout, netlist, SPEF, SDC, reports, release notes, checklist, etc.) submitted to the foundry.

Q2. Why is technohlay important ?

→ It provides extracted RC parasitics for accurate post-layout STA and sign-off.

Q3. What is the role of release notes ?

→ It documents design overview, assumptions, ECOs, waivers and any special instructions for the foundry.

Author : Konda Tharun Vivekananda

TAPEOUT RIGHT ! SILICON SUCCESS !!

Post - PD GDS II Generation and TapeOut

26. TAPEOUT FUNDAMENTALS - INTRODUCTION

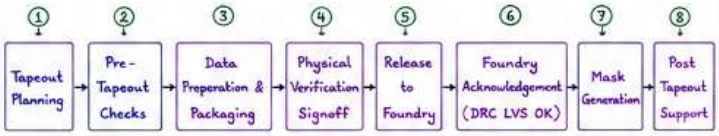
- TapeOut is the final step in ASIC design where the verified physical layout database is handed over to the foundry for fabrication.
- All design data must be complete, consistent, sign-off clean and packed as per foundry requirement.
- Any mistake after tapeout is extremely expensive and time consuming.

> KEY POINT
> Tapeout is not just sending GDS. It is a complete, clean and well documented data package transfer to the foundry.

27. EVOLUTION OF TAPEOUT

PERIOD	TECHNOLOGY	TAPEOUT FORMAT	CHARACTERISTICS
1980's	Micron (>1um)	Mask drawings, Paper tape	Manual mask generation, Slow, error prone.
1990's	Deep Submicron (0.8um - 0.35um)	GDS II	First digital data format, large file size.
2000's	Sub-100nm (180nm - 65nm)	GDS II + Compression	Need for compression, hierarchical data.
2010's	Advanced Nodes (45nm - 16nm)	OASIS	High compression, industry standard.
2020's	FinFET / EUV (<10nm)	OASIS + Incremental	Huge designs, eco flows, incremental tapeout.

28. TAPEOUT MILESTONES (STAGES)



① Tapeout Planning : Define schedule, freeze date, resources, ECO strategy.
② Pre-Tapeout Checks : Run all checks, fix violations, review metrics.
③ Data Preparation : Final data collection, naming, packaging, encryption.
④ Physical Verification Signoff : DRC, LVS, ERC, Antenna, etc signoff.
⑤ Release to Foundry : Submit the complete data package.
⑥ Foundry Acknowledgement : Foundry runs internal checks and confirms.
⑦ Mask Generation : Foundry generates masks for fabrication.
⑧ Post Tapeout Support : Support foundry queries, ECOs, metal fixes, etc.

29. PHYSICAL VERIFICATION SIGNOFF

✓ All verification steps must be sign-off clean before tapeout.
✓ Typical signoff checklist :

- → DRC (Design Rule Check)
- → LVS (Layout vs Schematic)
- → Density Check
- → Device Check
- → Stitching / Multi Patterning
- → ESD Check
- → ERC (Electrical Rule Check)
- → Antenna Check
- → Metal Fill / CMP Check
- → Well / Tap / DFM Rules
- → Architecture Checks
- → Seal Ring / Die Boundary

✓ All violations = 0 (zero critical / zero waived) unless approved.
✓ Signoff reports are generated and archived.

> SIGNOFF STATUS
> DRC : CLEAN LVS : CLEAN ERC : CLEAN ANTENNA : CLEAN
> DENSITY : CLEAN DFM : CLEAN OVERALL : READY FOR TAPEOUT

30. FOUNDRY HANDOFF

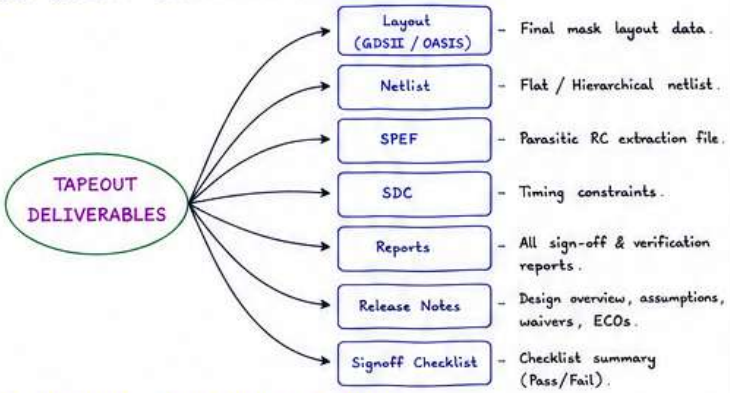
* After signoff, the final daataas package is handed to the foundry.
* Foundry runs their internal verification (DRC / LVS / Antenna / Density).
* If foundry reports any issue → must be fixed and re-submitted.
* Foundry issues "Data Acceptance" / "In-Review" / "Approved".
* Only after "Approved" → masks will be generated.

> FOUNDRY DATA ACCEPTANCE STATES
> In-Review → Issues Found → Fixed & Resubmitted → Approved

31. POST TAPEOUT SUPPORT

* Foundry may raise queries during data review.
* Engineering team supports by providing missing files or clarifications.
* If any metal fix / DRC fix / Density fix is required by foundry, then ECO flow is followed.
* Mask house generates masks after final approval.
* After wafer start → test chips come → bring up → characterization.

32. TAPEOUT DELIVERABLES OVERVIEW



33. TAPEOUT DELIVERABLES - DETAILS

DELIVERABLE	DESCRIPTION	PURPOSE	FORMAT / EXTENSION
1. Layout (GDSII / OASIS)	Final geometric database with all layers.	Mask fabrication at foundry.	.gds / .oas
2. Netlist	Gate-level netlist. (top & flattened)	LVS and documentation.	.v / .sp / .sv
3. SPEF	Parasitic RC extraction file (post-layout).	Post-layout STA, SI, IR/EM analysis.	.spef
4. SDC	Synopsys Design Constraints.	Define timing, I/O, power, false paths.	.sdc
5. Reports	All sign-off reports from tools.	Proof of verification and signoff.	.rpt / .log / .html
6. Release Notes	Design overview, tool versions, ECOs, waivers, special instructions.	Reference for foundry and future teams.	.pdf / .doc / .txt
7. Signoff Checklist	Summary checklist of all verifications.	Quick status (Pass/Fail).	.xls / .xlsx / .pdf
8. Layer Map	Mapping of foundry layers.	Layer interpretation.	.layermap
9. DFT / Scan Data	Scan chain info, test mode data.	Manufacturing test support.	various
10. Power Intent	UPF / CPF file.	Power domain information.	.upf / .cpf
11. Library / Liberty	Liberty timing / power libraries used.	Reference for foundry / STA.	.lib
12. Other Files	LEF, DEF, CDL, ILM, Antenna rules etc.	Additional data as required.	various

> KEY POINT
> ✓ Complete and clean deliverables ensure smooth tapeout.
> ✓ Missing or wrong file → tapeout delay / rejection.
> ✓ Always follow foundry tapeout manual strictly.

34. SUMMARY / REVISION POINTS

- Tapeout is the final transfer of verified design to the foundry.
- All data must be signoff clean and packaged correctly.
- Key milestones: planning → checks → packaging → signoff → handoff.
- Deliverables include layout, netlist, SPEF, SDC, reports, notes, checklist.
- Verification signoff is mandatory before tapeout.
- Foundry handoff requires approval before masks are generated.
- Post tapeout support is required for queries, ECOs and bring-up.

35. VIVA / INTERVIEW POINTS

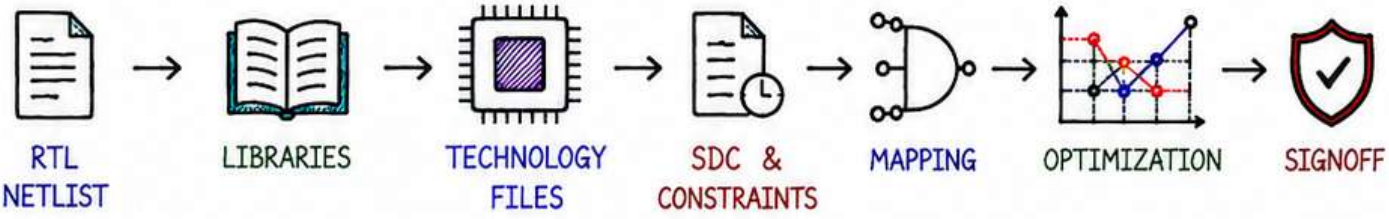
Q1. What is tapeout?
→ It is the final handoff of signoff clean design data to the foundry.

Q2. What are key tapeout deliverables?
→ Layout (GDS/OASIS), Netlist, SPEF, SDC, Reports, Notes, Checklist.

Q3. Why is verification signoff important?
→ To ensure the layout is correct and manufacturable with zero violations.

Author : Konda Tharun Vivekananda

CLEAN SIGNOFF → CONFIDENT TAPEOUT → SUCCESSFUL SILICON !



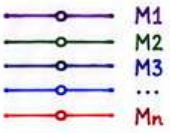
GDS II

THANK YOU!

Thank you for your time, interest and support in exploring GDS II.
Hope these notes help you understand and work confidently with GDS II files.

GDS II AT A GLANCE

- Industry standard layout data format.
- Hierarchical, precise and manufacturable.
- Used for fabrication, verification and analysis.



KEY TAKEAWAYS

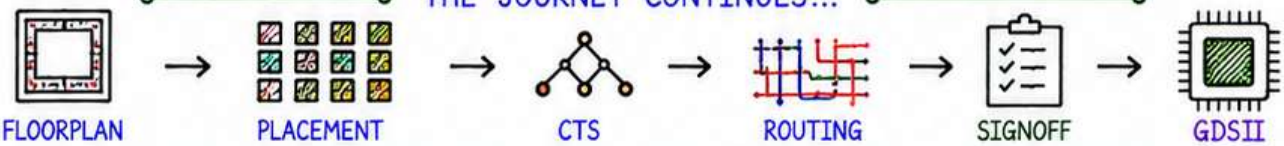
- ☆ GDS II is the final physical representation of the chip.
- ☆ Accurate stream-out and verification ensure clean signoff.
- ☆ Follows foundry rules and data requirements.
- ☆ Essential for fabrication, yield and reliability.
- ☆ Quality GDS = Quality Silicon.

WHAT GOOD GDS II DELIVERS

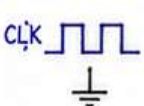
- ✓ Clean DRC and LVS
- ✓ Reliable Manufacturing
- ✓ Better Yield
- ✓ Faster Time to Market
- ✓ Accurate Analysis
- ✓ Robust Chip Performance
- ✓ Happy Foundry ☺

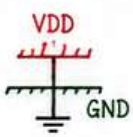
"A clean, verified GDS II is the bridge between design and silicon."

THE JOURNEY CONTINUES...



Keep learning, keep building, keep innovating!
Wishing you the best for your journey in Physical Design!





Author :
Konda Tharun Vivekananda